✅ Q1 (a)

Question:

Explain bias, variance, and bias–variance trade-off in machine learning with the help of examples.

Answer:

1️⃣ Bias

Bias is the error caused due to overly simple assumptions in the learning algorithm.

- High bias → Model is too simple
- Causes **underfitting**

Example:

If we fit a straight line to data that follows a curve.

Actual relationship:

$$y = x^2$$

Model used:

$$y = ax + b$$

The model cannot capture the curve → High Bias.

Real-world Example:

If we predict house prices using only 1 feature (size) and ignore location, rooms, age → model will be too simple → high bias.

2️⃣ Variance

Variance is the error caused when the model is too sensitive to training data.

- High variance → Model memorizes training data
- Causes **overfitting**

Example:

Using a very high-degree polynomial that passes through every training point exactly.

Real-world Example:

A stock prediction model that perfectly fits past data but fails for future prices.

3️⃣ Bias–Variance Tradeoff

There is always a tradeoff:

Model Complexity	Bias	Variance
Simple model	High	Low
Complex model	Low	High

Goal:
Minimize total error.

$$TotalError = Bias^2 + Variance + IrreducibleError$$

Conclusion:

A good model balances bias and variance to generalize well on unseen data.

✅ Q1 (b)

Question:

Compare how logistic regression is different from linear regression. Explain regularization with an example.

Answer:

Difference Between Linear and Logistic Regression

Feature	Linear Regression	Logistic Regression
Used For	Predict continuous values	Classification
Output	Any real value	Probability (0–1)
Function	y = mx + c	Sigmoid function
Cost Function	MSE	Log Loss

Linear Regression

Used for predicting continuous values.

Example:

- Predict house price
- Predict salary
- Predict temperature

Equation:

$$y = \theta_0 + \theta_1 x$$

Logistic Regression

Used for classification problems.

Example:

- Spam or Not Spam
- Disease or No Disease
- Pass or Fail

Uses Sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Output is probability.

Regularization

Regularization prevents overfitting by adding penalty to large weights.

Modified cost function:

$$J(\theta) = Loss + \lambda \sum \theta^2$$

Where:

- λ = regularization parameter

Real-world Example:

In medical diagnosis:
If the model uses too many features (blood tests, reports, noise data), it may overfit.

Regularization reduces weight of unnecessary features.

Types:

- L1 (Lasso)
- L2 (Ridge)

Conclusion:

Linear regression → best for continuous prediction.
Logistic regression → best for classification.
Regularization → prevents overfitting.

✅ Q2 (a)

Question:

Explain three different validation techniques namely Hold-out, K-Fold cross validation and Leave-one-out. Discuss their pros and cons.

Answer:

Validation is used to evaluate model performance.

1️⃣ Hold-Out Method

Data is split into:

- Training set
- Testing set

Example:
70% training
30% testing

Real-world Use:

Used in quick ML projects where dataset is large.

Pros:
✔ Simple
✔ Fast

Cons:
✖ Result depends on how data is split

2️⃣ K-Fold Cross Validation

Dataset is divided into K equal parts.

Process:

- Train K times
- Each fold used once as test set
- Final accuracy = average

Example:
K = 5

Used in:
Model selection and hyperparameter tuning.

Pros:
✔ More reliable
✔ Uses entire dataset

Cons:
✖ More computational cost

3️⃣ Leave-One-Out Cross Validation (LOOCV)

Special case of K-fold where:

$$K = N$$

If dataset has 100 samples:
Train 100 times.

Used when:
Dataset is very small.

Pros:
✔️ Very accurate
✔️ Maximum data usage

Cons:
✖ Very slow

Conclusion:

- Large dataset → Hold-out
- Medium dataset → K-Fold
- Small dataset → LOOCV

✅ Q2 (b)

Question:

What is regularization, and why is it used in machine learning? How does regularization help in preventing overfitting?

Answer:

Regularization is a technique used to reduce overfitting by adding penalty to large weights.

Modified cost function:

$$J = Loss + \lambda \mid\mid \theta \mid\mid^2$$

Why Regularization is Used?

When model becomes too complex:

- Fits training data perfectly
- Performs poorly on test data

Regularization makes model simpler.

How It Prevents Overfitting?

Large weights → complex decision boundary
Regularization → reduces weight size
Simpler model → better generalization

Types:

L1 Regularization (Lasso)

Adds:

$$\lambda \sum | \theta |$$

- Creates sparse model
- Feature selection

L2 Regularization (Ridge)

Adds:

$$\lambda \sum \theta^2$$

- Shrinks weights smoothly

Real-world Example:

In image classification:
If we use too many features, model may memorize images.

Regularization helps model focus on important features only.

✅ Q3

Question:

Consider the following training set having three attributes x_1, x_2, x_3:

X1	X2	X3	Class
-1	1	1	Negative
5	1	0	Positive
0	2	1	Negative
-4	-1	0	Negative
0	0	1	Positive
9	2	1	Positive
2	2	0	Negative

Using KNN (K=3), classify:

i) (1,1,2)
ii) (2,1,1)
iii) (0,0,0)

Solution:

We use **Euclidean Distance**:

$$d = \sqrt{(x_1 - a)^2 + (x_2 - b)^2 + (x_3 - c)^2}$$

🔷 i) Classify (1,1,2)

Compute distances with all points (show nearest 3 only):

Closest neighbors:

- (0,2,1) → Negative

- (-1,1,1) → Negative
- (0,0,1) → Positive

Majority = 2 Negative, 1 Positive

✅ **Final Class = Negative**

🔷 ii) Classify (2,1,1)

Nearest neighbors:

- (5,1,0) → Positive
- (0,2,1) → Negative
- (0,0,1) → Positive

Majority = 2 Positive

✅ **Final Class = Positive**

🔷 iii) Classify (0,0,0)

Nearest neighbors:

- (0,0,1) → Positive
- (-1,1,1) → Negative
- (2,2,0) → Negative

Majority = 2 Negative

✅ **Final Class = Negative**

Real-world Applications of KNN:

- Medical diagnosis
- Face recognition
- Recommendation systems
- Fraud detection

✅ Q4

Q4 (a)

Question:

What is Locally Weighted Regression? How does it differ from KNN?

Answer:

Locally Weighted Regression (LWR)

It is a regression algorithm that gives more weight to nearby points.

Weight formula:

$$w(i) = e^{-\frac{(x-x_i)^2}{2\tau^2}}$$

Nearby points → high weight
Far points → low weight

Difference Between LWR and KNN

Feature	LWR	KNN
Output	Continuous	Class label
Type	Regression	Classification/Regression
Weight	Uses weight function	Equal vote

Real-world Use:

- Predicting house price based on nearby houses
- Weather prediction based on nearby locations

Q4 (b)

Question:

A linear SVM separates two classes using:

$$3x - 4y - 10 = 0$$

Find the margin.

Solution:

General form:

$$ax + by + c = 0$$

Here:
a = 3
b = -4

Margin formula:

$$Margin = \frac{2}{\sqrt{a^2 + b^2}}$$
$$= \frac{2}{\sqrt{3^2 + (-4)^2}}$$
$$= \frac{2}{\sqrt{9 + 16}}$$
$$= \frac{2}{5}$$
$$= 0.4$$

✅ **Margin = 0.4**

Real-world Use of SVM:

- Face detection
- Text classification
- Image recognition
- Cancer detection

✅ Q5

Q5 (a)

Question:

What is Boosting? Explain with decision trees.

Answer:

Boosting is an ensemble technique that combines multiple weak learners to create a strong learner.

AdaBoost Working:

1. Train first decision tree.
2. Increase weight of misclassified points.
3. Train second tree.
4. Repeat.
5. Combine predictions using weighted voting.

Why Boosting Works?

It focuses more on difficult samples.

Real-world Applications:

- Fraud detection
- Spam filtering
- Credit risk analysis
- Face detection

Q5 (b)

Given linear model:

$$Y = 5X + 2$$

Dataset:

X	Y
2	12
4	22
6	32
8	42

Step 1: Predicted Values

For X=2 → 12
For X=4 → 22
For X=6 → 32
For X=8 → 42

Step 2: Calculate MSE

$$MSE = \frac{1}{n}\sum(y - \hat{y})^2$$

All predicted values match actual.

$$MSE = 0$$